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Microbial community structure reveals how microaeration improves fermentation during anaerobic co-digestion of brown water and food waste

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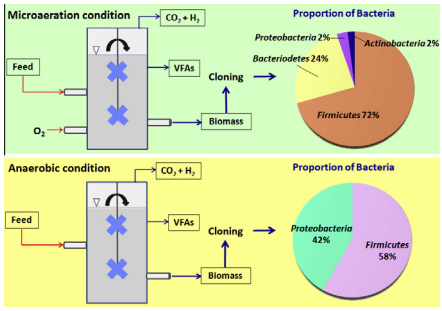
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HIGHLIGHTS

- Microaeration gave rise to a significantly more diverse bacterial population.
- Higher proportion of clones affiliated to *Firmicutes* in microaeration reactor.
- Microaeration led to a shift in fermentation production pattern.
- Microaeration enhanced fermentation during co-digestion of BW and FW.

GRAPHICAL ABSTRACT



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ABSTRACT

The purpose of this study was to investigate the impact of microaeration on the fermentation process during anaerobic co-digestion of brown water (BW) and food waste (FW). This was achieved by daily monitoring of reactor performance and the determination of its bacterial consortium towards the end of the study. Molecular cloning and sequencing results revealed that bacteria within phyla *Firmicutes* and *Bacteriodetes* represented the dominant phylogenetic group. As compared to anaerobic conditions, the fermentation of BW and FW under microaeration conditions gave rise to a significantly more diverse bacterial population and higher proportion of bacterial clones affiliated to the phylum *Firmicutes*. The acidogenic reactor was therefore able to metabolize a greater variety of substrates leading to higher hydrolysis rates as compared to the anaerobic reactor. Other than enhanced fermentation, microaeration also led to a shift in fermentation production pattern where acetic acid was metabolized for the synthesis of butyric acid.

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1. Introduction

Anaerobic digestion (AD) refers to the fermentation process that produces biogas (composed of mainly methane and carbon dioxide) from the degradation of organic material. Due to the production of useful energy in the form of biogas, AD has been widely applied for the treatment of organic waste such as brown water

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(BW) and food waste (FW) (Rajagopal et al., 2013; Curry and Pillay, 2012; Zeeman et al., 2008). AD is a complex degradation pathway that proceeds in four successive stages, namely hydrolysis, acidogenesis, acetogenesis and methanogenesis. Acid-forming bacteria carry out hydrolysis, acidogenesis and acetogenesis, while methane-forming archaea produce biogas during methanogenesis. Due to the different nutritional needs of the microorganisms involved in each stage, the physical separation of acid- and methane-forming microorganisms in two separate reactors was first proposed by Pohland and Ghosh (1971) to provide optimum environmental conditions for each group of microorganisms. Recent studies have indeed shown that two-phase systems could lead to enhanced stability and control of the overall AD process (Lim et al., 2013; Demirel and Yenigun, 2002).

Though the AD technology existed for more than 100 years, there are still unresolved challenges faced by AD operators, thus limiting the implementation of more AD plants. In view of current global concerns over environmental sustainability, AD is regarded as a promising process due to its potential in renewable energy generation and waste stabilization. However, one of the main problems faced by AD operators is its inherent instability since AD is alarmingly sensitive to changes in operation and feed conditions. Accidental or unavoidable oxygen loading is one aspect of this problem. Theoretically, the AD process will be inhibited when exposed to oxygen. However, several studies have shown that partial aeration did not cause any inhibition, but enhanced AD performance instead (Botheju and Bakke, 2011). According to Botheju and Bakke (2011), the presence of oxygen led to a higher yield and population of facultative acidogens and therefore higher amount of enzymes excreted. It was hypothesized that more acidogenic biomass leads to more hydrolysis, since hydrolysis is carried out by the extracellular enzymes excreted by acidogens.

The term microaeration refers to the controlled introduction of small amounts of oxygen into an anaerobic biochemical process to enable both anaerobic and aerobic biological activities to occur within a single bioreactor. The term microaeration will be used in this study to describe the conditions of adding oxygen to the acidogenic reactor. The study by Rolfe et al. (1978) was among the first few that investigated the oxygen tolerance level of anaerobic bacteria. Zitomer and Shrout (1998) subsequently reported that oxygen addition did not inhibit the growth of methanogens, but increased their initial activity. More recently, several studies reported advantages of microaeration in terms of higher degree of solubilization and acidification of organic matter (Xu et al., 2014a,b; Lim and Wang, 2013; Díaz et al., 2011a; Jagadabhi et al., 2010). Therefore, microaeration has been regarded as a potential pre-treatment method for improving the hydrolysis stage during the AD process. Another reported benefit of microaeration was the cleaning of biogas by removing more than 99% hydrogen sulfide (H_2S) (Díaz et al., 2011b; Tang et al., 2004). As compared to the other chemical and physical pre-treatment methods or biological processes for the desulphurization of biogas, microaeration of AD system has a relatively smaller footprint and require lower investment costs as well as small modification to the existing process (Ramos et al., 2014).

An earlier study on the anaerobic co-digestion of BW and FW (Lim et al., 2013) reported the unexpected predominance of aerobic bacteria species – *Acetobacter peroxydans* in the acidogenic reactor of a two-phase continuously stirred tank reactor (CSTR). As the acidogenic reactor was operated under anaerobic conditions, the predominance of an aerobic bacteria species suggested the reactor might be unknowingly exposed to partial aeration. Despite the predominance of *A. peroxydans*, the AD system achieved high degrees of COD solubilization and VFA production. This study illustrated that in case of accidental or unavoidable oxygen loading, the fermentation process of AD was not compromised.

In view of the tendency for accidental or unavoidable oxygen loading, as well as the benefit of microaeration in terms of enhanced fermentation and desulphurization of biogas, it is important to understand how AD systems respond to small amounts of oxygen stress. Determining the microbial diversity of reactors will provide more insights on the changes in the biochemical processes due to microaeration conditions. However, information on the microbiology of anaerobic digesters operated under microaeration conditions is limited (Ramos et al., 2014; Zhou et al., 2007; Tang et al., 2004).

Tang et al. (2004) reported that microaeration led to a decrease in *Methanosarcina* and increase in *Methanoculleus* populations while Zhou et al. (2007) found that limited aeration caused the predominant microorganisms to change from rod-shape to cocci-shaped methanogens in the UASB reactor. The DGGE analysis carried out by Ramos et al. (2014) showed that oxygen affected the richness, evenness and structure of the bacterial and the archaeal communities in the long term. These studies mainly discussed the effect of microaeration on the archaeal populations and very little is known about the bacterial community shifts due to oxygen. Since one of the main benefits of microaeration was reported to be enhanced hydrolysis, it is essential to have a more detailed understanding of how microaeration affects the bacterial population. Therefore, the objective of this study was to investigate how microaeration affected the fermentation process in the anaerobic co-digestion of BW and FW. This aim was achieved by determining the bacterial community structure of the acidogenic reactor of a two-phase CSTR and correlating the microbial structure to the reactor's performance.

2. Methods

2.1. Experimental set-up

The feed for this study consisted of a mixture of 56.25 g blended food waste (FW) and 0.75 L brown water (BW) with an average pH of 5.96 ± 0.22. The FW/BW mixture was prepared and fed daily to the acidogenic reactor of a two-phase CSTR system, in batch mode. The working volume of the reactor was 3 L and the contents were mixed continuously at 80 rpm by an overhead mechanical stirrer. The reactor was initially inoculated with sludge collected from another acidogenic reactor treating BW and FW at 35 °C for more than one year. With hydraulic retention time (HRT) of 4 days, the organic loading rate (OLR) for the acidogenic reactor was maintained at 5.15 ± 0.44 g-VS/L/d in this study. As shown in Table 1, the study consisted of three operating conditions. The reactor was operated under anaerobic conditions (AN) from week 1 to 6, low microaeration conditions (MA1) from week 7 to 13, and higher microaeration conditions (MA2) from week 14 to 20. Oxygen was added to the liquid part of the reactor at a rate of 3 mL/min daily one to two hours after feeding.

2.2. Chemical analysis

The reactor performance was monitored daily for pH, oxidation reduction potential (ORP) and biogas while total solids (TS), volatile solids (VS), chemical oxygen demand (COD), volatile fatty acid (VFA) and ammonia (NH_3-N) were monitored weekly. All analysis were carried out in duplicates.

pH value was measured using a compact titrator (Mettler Toledo) equipped with a pH probe (Mettler Toledo DGi 115-SC). ORP was determined using a platinum ORP combination electrode (Fisher Scientific Accumet® Ca^+ Model 13-620-81). Biogas was collected and stored in a Tedlar gas sampling bag (Sigma–Aldrich, USA) and its volume was monitored daily using a rotary displacement

Table 1
Average values of parameters for acidogenic reactor during AN, MA1 and MA2 conditions.

Condition	Week	pH	ORP (mV)	NH_3-N (mg/L)	TVFA (mg-COD/L)	H-Ac	H-Pr	H-Bu	SCOD (mg/L)
AN	1–6	4.00	−97	96	7789	1560	918	2440	17,105
MA1 (5 mL-O_2/L_R/d)	7–13	4.12	−100	110	11,143	1608	1715	4199	17,529
MA2 (7 mL-O_2/L_R/d)	14–20	4.07	−172	99	10,281	1337	1541	4482	17,775

meter. The composition of biogas (methane, carbon dioxide and nitrogen contents) was analyzed by gas chromatograph (Agilent Technologies 7890 A, USA) equipped with a thermal conductivity detector (TCD). TS and VS were analysed according to the Standard Methods (APHA, 1998) while total and soluble COD were measured using COD digestion vials (Hach Chemical) and a spectrophotometer (DR/2800, Hach). Soluble COD was measured using the supernatant of samples after centrifugation (KUBOTA 3700, Japan) at 12,000 rpm for 10 min. The supernatant of samples were filtered through Membrane Solutions 0.45 μm cellulose acetate membrane filters for the analysis of VFAs using a gas chromatograph (Agilent Technologies 7890A, USA), equipped with a flame ionization detector (FID) and a DB-FFAP (Agilent Technologies, USA) column (30 m × 0.32 mm × 0.50 μm).

2.3. Microbial analysis

To further understand the system, biomass were collected towards the end of each condition of the study for microbial community characterization. Genomic DNA in the biomass samples were extracted using chemical lysis and phenol–chloroform–isoamyl alcohol (25:24:1, v:v:v) purification protocol. Primer set 8F (5′-AGAGTTTGATYMTGGCTC-3′) and 1490R (5′-GGTTACCTTGTTACGACTT-3′) was used to amplify Bacterial 16S rRNA gene from the total-community DNA. The thermal program used for amplification of 16S rRNA gene was as follows: hotstart 94 °C for 3 min, 30 cycles of denaturation (30 s at 94 °C), annealing (30 s at 54 °C), extension (45 s at 72 °C) and a final extension at 72 °C for 5 min.

Molecular cloning and sequencing was used to determine the microbial consortium in the reactor. TOPO TA cloning kit (Invitrogen, CA) was used for clone library construction according to the manufacturer's instructions. Approximately 100 clones were randomly selected for members in the domain Bacteria (amplified by primer set 8F and 1490R). The amplified DNA insert was then PCR amplified with a vector-specific primer set (i.e., M13F and M13R). The 16S rRNA gene fragments were screened by restriction fragment length polymorphism (RFLP) to further remove the possible redundant clones. This was followed by the separate digestion of M13-PCR products with tetramer restriction enzymes *Msp*I and *Rsa*I (New England BioLabs, UK). Electrophoresis was carried out to separate the digestion products in a 3% agarose gel. The gels were then visualized using the FireReader gel documentation (UVItec, Cambridge, UK) after staining with Gelred (Invitrogen, CA). Unique RFLP patterns were defined as a unique sequence type of operational taxonomy unit (OTU).

The 16S rRNA gene of the representative clones with different RFLP patterns were sequenced, by Axil Scientific Sequencing (Singapore), to determine their phylogenetic affiliation. Nearly full-length 16S rRNA gene sequences of representative clones were compared to available rRNA gene sequences in GenBank using the NCBI BLAST program. Chimeric artifacts were determined using DECIPHER (Wright et al., 2012) and phylogenetic trees were constructed with MEGA5 program using the remaining 61 bacterial clone sequences after removing chimeric sequences. The Jukes–Cantor correction was used for distance matrix analyses and the trees were constructed using the Neighbor-joining method. Bacterial 16S rRNA partial sequences obtained in this study were deposited in the nucleotide Genbank database, under the accession numbers: KJ907449–KJ907466.

3. Results and discussion

3.1. Digester performance

3.1.1. Oxidation reduction potential (ORP) levels

ORP level was measured daily to monitor the level of oxidants in the reactor. In fully anaerobic systems, ORP values vary between −200 and −300 mV while in aerobic systems, ORP increases to above 50 mV (Henze, 2008). As shown in Fig. 1a, ORP levels dropped to negative values within 1–2 weeks from the start of the study. Throughout the AN condition, the average ORP levels decreased from 226 mV (in week 1) to −168 mV (in week 6). According to Xu et al. (2014a,b), the optimum ORP range for acid fermentation was between −100 mV and −250 mV. In this study,

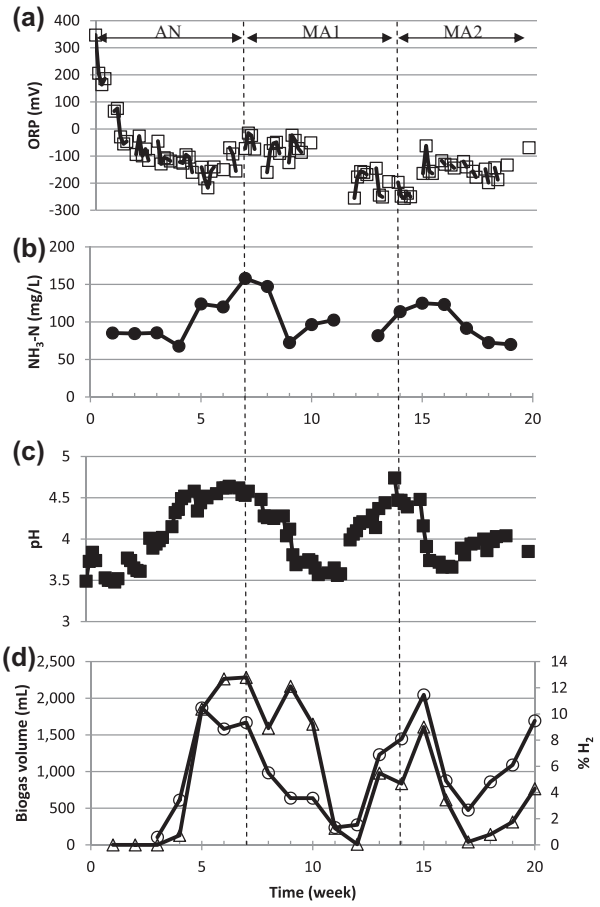


Fig. 1. Performance of acidogenic reactor: (a) –□– ORP, (b) –●– NH_3-N, (c) –■– pH, (d) –○– biogas volume and –△– %H_2.

the acidogenic reactor reached redox conditions favoring fermentation within 2 weeks from the start of the study.

Operation changed from anaerobic (AN) to microaeration (MA1) conditions during week 7, where 5 mL-O_2/L_R/d was added to the acidogenic reactor. Due to the introduction of oxygen into the acidogenic reactor, ORP levels became less negative. Although ORP values were still within the negative range, the rising ORP levels indicated that the microorganisms did not manage to adapt well to the sudden supply of oxidants. ORP levels only recovered to between −157 mV and −255 mV (i.e., the range favorable for fermentation) towards the end of MA1 (in week 12). This could be attributed to the acclimatization of microbial community towards microaeration conditions in the acidogenic reactor. The ORP trend suggested that the growth of microorganisms able to survive oxidative stress were stimulated during MA1, which led to the lowering of ORP levels towards the end of MA1.

The oxygen dosage increased from 5 mL-O_2/L_R/d to 7 mL-O_2/L_R/d from week 14 onwards (MA2 conditions). Due to the higher oxidant level present in the acidogenic reactor, ORP levels increased slightly but quickly stabilized at levels between −117 mV and −256 mV after one week. This suggested that the bacterial groups cultivated during MA1 were better able to tolerate or consume additional oxygen added and adapted to higher levels of oxidants quickly. According to Table 1, the average ORP levels during AN, MA1 and MA2 conditions were −97 mV, −100 mV and −172 mV, respectively. Although oxygen was added to the acidogenic reactor during MA1 and MA2, the average ORP values during each period became increasingly negative with higher levels of oxygen added. After an adaptation period of approximately 6 weeks (during weeks 7 to 12), the bacterial population in the acidogenic reactor was able to effectively consume oxygen and maintain a reducing environment such that fermentation could occur despite higher amounts of oxygen introduced.

3.1.2. NH_3-N, pH and biogas production

According to Table 1, the average NH_3-N levels during AN, MA1 and MA2 were 96 mg/L, 110 mg/L and 99 mg/L, respectively. As shown in Fig. 1b, microaeration led to increasing NH_3-N levels, and were highest during weeks 7 and 15 – the first one to two weeks after introducing MA1 and MA2 conditions, respectively.

During anaerobic degradation, proteins are solubilized into amino acids, which are then degraded into NH_3 and VFAs such as acetic acid (H-Ac), propionic acid (H-Pr) and butyric acid (H-Bu). Therefore, the spikes in NH_3-N levels observed during increasing dosage of oxidants suggested that microaeration led to a greater extent of proteins degradation. NH_3-N levels were highest during week 7 (158 mg/L) and week 15 (125 mg/L) but quickly returned to lower levels after one and two weeks of MA1 and MA2, respectively. The drop in NH_3-N levels during weeks 8 and 16 could mean the microorganisms quickly adapted to increased levels of oxidants.

pH levels ranged between 3.5 and 4.6 in this study (Fig. 1c). As shown in Table 1, pH levels increased from 4.00 to 4.12 and dropped from 4.12 to 4.07 as conditions changed from AN to MA1 and from MA1 to MA2, respectively. On top of that, Fig. 1b and c show that pH levels followed the same trend as NH_3-N levels. pH levels rose with increasing levels of NH_3-N and pH levels dropped as NH_3-N levels decreased.

According to the Stickland reaction, NH_3 is released during the acidogenic conversion of amino acids. Since NH_3 is a weak base that will partially react with water to produce an equilibrium concentration of hydroxyl anion (OH^-), increased levels of NH_3-N led to a pH rise. Biogas produced from the fermentation process was measured daily. As shown in Fig. 1d, the biogas production and its composition of H_2 also followed the same trend as pH.

3.1.3. Soluble COD and VFA

According to Table 1, the average levels of soluble COD and TVFA were higher during MA1 and MA2, as compared to AN conditions. This indicated that microaeration conditions led to a greater extent of solubilization and acidification as compared to anaerobic conditions. The increase in TVFA levels could also be attributed to the enhanced degradation of protein since the degradation of proteins produces VFAs such as H-Ac, H-Pr and H-Bu. In addition, Ding (2010) also found that protein degradation promoted the activity of acidogenic microorganisms by providing readily available organic nitrogen in the form of soluble protein and amino acids. Therefore, enhanced protein degradation during MA conditions contributed to the rise in VFA production.

Among the VFAs produced, 60% were composed of H-Ac and H-Bu, where H-Bu accounted for up to 42%. In this study, the change in TVFA levels is mainly influenced by H-Bu levels. As shown in Fig. 2, the increase and decrease in TVFA levels follow the same trend as that of H-Bu levels. High concentrations of H-Ac and H-Bu was also reported in other studies that utilized food waste as substrate for anaerobic digestion (Xu et al., 2014a,b; Han and Shin, 2004). According to Xu et al. (2014a,b), H-Bu and H-Ac are regarded as preferred precursors for CH_4 production. Therefore, conversion of organic matter into H-Ac and H-Bu in the hydrolytic-acidogenic stage will improve the overall energy yield and increase the process efficiency.

With higher dosage of oxidants, pH, TVFA and H-Bu levels increased. On the other hand, H-Ac levels increased during MA1 but decreased during MA2. As shown in Fig. 2, H-Ac levels dropped as H-Bu levels increased. While %H-Ac of TVFA dropped from 38% to 16% to 14%, %H-Bu increased from 22% to 37% to 42% during AN, MA1 and MA2 conditions, respectively. This suggested that microaeration not only accelerated fermentation, but also had an impact on the fermentation product patterns (i.e., shifted the metabolism from H-Ac production to H-Bu production).

The high levels of H-Bu reported in this study was likely due to the presence of butyrate-producing bacteria. H-Bu production generally proceeds via two distinct pathways, via butyrate kinase or butyryl coenzyme A (CoA)–acetyl CoA transferase (Pryde et al., 2002; Duncan et al., 2002). The study by Pryde et al. (2002) reported that the butyryl CoA–acetyl CoA transferase pathway was more dominant in the human bacterial flora. Another study also found that 50% of the butyrate-producing isolates were net H-Ac consumers during growth, suggesting the preference of the butyryl CoA–acetyl CoA transferase pathway (Morrison et al., 2006). This was similar to the results reported in a mixed human faecal incubation study which found that a significant amount of carbon in H-Bu was contributed by free H-Ac (Barcenilla et al., 2000). Since the butyryl CoA–acetyl CoA transferase pathway is more dominant in the

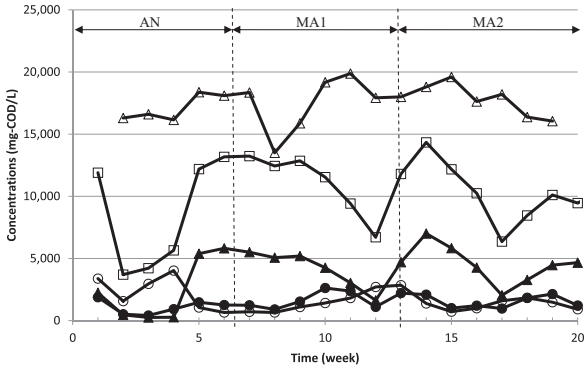


Fig. 2. Performance of acidogenic reactor: –△– SCOD, –□–TVFA, –○– H-Ac, –●–H-Pr and –▲– H-Bu.

human bacterial flora, the high levels of H-Bu and low levels of H-Ac during microaeration conditions was likely due to the metabolism of H-Ac by the butyrate-producing bacteria, for the synthesis of H-Bu.

It was also observed that H-Ac levels decreased whenever H_2 production increased. During acetogenesis, hydrogen-producing acetogenic bacteria convert higher VFAs such as H-Pr and H-Bu into H_2, CO_2 and H-Ac. H_2 production by acetogens is generally energetically unfavorable due to high free energy requirements. Without the combination of H_2-consuming bacteria in the acidogenic reactor, conditions were unfavorable for the production of H-Ac, therefore contributing to the drop in H-Ac levels.

3.2. Bacterial community characterization

3.2.1. Overview

As shown in Fig. 3, the bacterial community structure of the acidogenic reactor was composed of phyla *Firmicutes*, *Bacteriodetes*, *Proteobacteria* and *Actinobacteria* in proportions of 72%, 24%, 2% and 2% of the bacterial clones, respectively. In total, 18 bacterial operational taxonomic units (OTU) were identified. Within the 18 OTU, 13 were classified as *Firmicutes*, 3 as *Bacteriodetes*, 1 as *Proteobacteria* and 1 as *Actinobacteria*. The most detected OTU (R1Bac_19), representing 33% of the total clones, was affiliated to *Megasphaera* species NMBHI-10 (HM990965) with 99% similarity. R1Bac_18 was the second most detected OTU accounting for 18% of the clones and was closely related to uncultured *Prevotella* species (DQ168844). The third most detected OTU was R1Bac_51 (10% of the bacterial clones) and was affiliated to *Lactobacillus amylovorus* GRL 1112 (NR_075048). R1Bac_19 and R1Bac_51 both belonged to phylum *Firmicutes*, which is known to produce cellulases, lipases, proteases and other extracellular enzymes (Levén et al., 2007). Therefore the predominance of *Firmicutes* (72% by cloning analysis) reflected the ability of the acidogenic reactor to metabolize a variety of substrates including protein, lipids, lignin, cellulose, sugars and amino acids, which are commonly found in FW. The predominance of *Firmicutes* was also reported in several studies that investigated the AD of FW (Lim et al., 2013; Shin et al., 2010; Tang et al., 2004). The next predominant phylum *Bacteriodetes* (24% by cloning analysis) represents a group of fermentative microorganism and is also a major microbial component of anaerobic reactors. In addition, *Bacteriodetes* is known to be proteolytic and play important roles in stabilizing semi-solid waste (Rivière et al., 2009; Kindaichi et al., 2004).

3.2.2. Dominant bacterial species

Bacteria within the phyla *Firmicutes* and *Bacteriodetes* represented the exclusive dominant phylogenetic group (96% by cloning analysis) in the acidogenic reactor, suggesting a major impact of these bacteria on the fermentation of BW and FW under microaerobic conditions. R1Bac_19 (33% of total clones) is closely related to the *Megasphaera* species NMBHI-10, whose genome was reported to code for enzymes essential for carbohydrate fermentation to yield H_2, CO_2, and VFAs such as H-Bu, H-Pr and H-Ac (Shetty et al., 2013).

The co-culture of lactic acid producing bacteria and *Megasphaera elsdenii* was previously shown to stimulate H-Bu production (Tsukahara et al., 2006). In this study, R1Bac_51 (10% of total

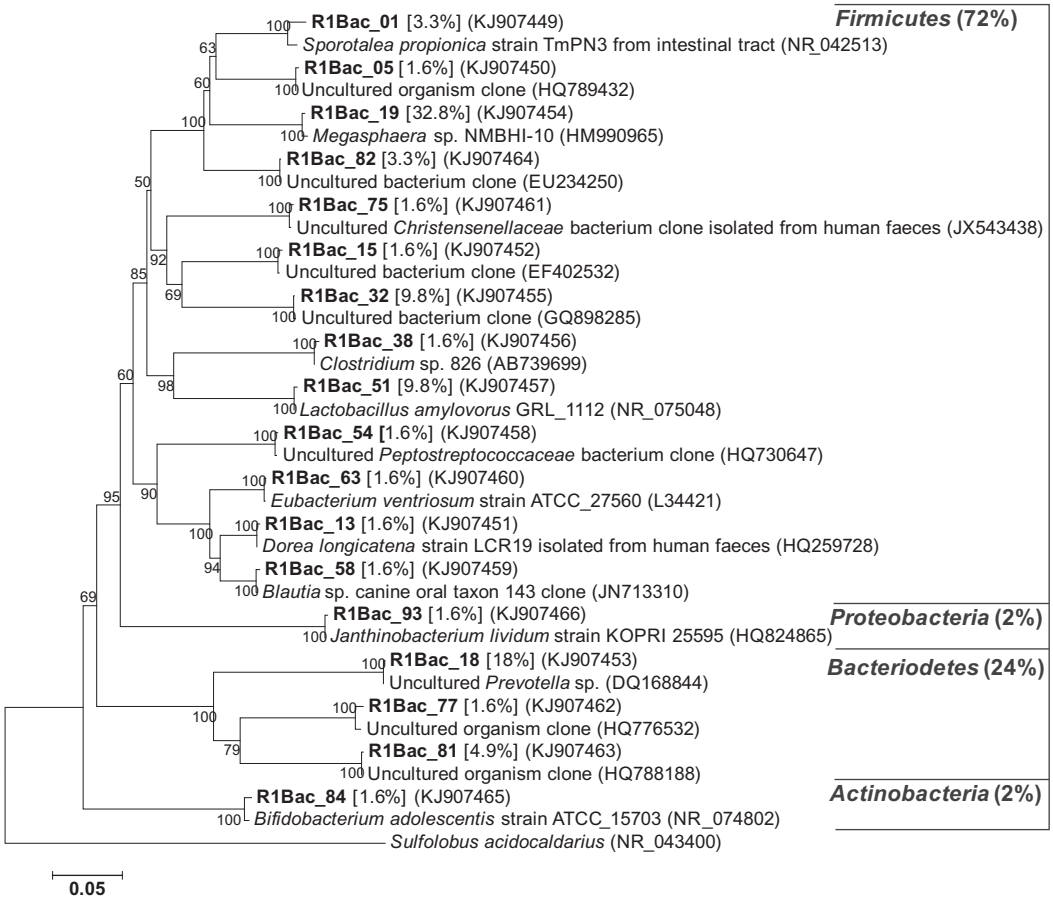


Fig. 3. Phylogenetic tree of 16S rRNA gene sequences constructed for bacterial clones from acidogenic reactor of two-phase CSTR. The 16S rRNA gene sequence of *Sulfolobus acidocaldarius* (NR_043400) was used as the outgroup.

bacterial clones) was affiliated to *L. amylovorus* – a lactate-producing bacteria possessing amylolytic activity. Therefore, the high levels of H-Bu in this study could also be due to the co-existence of and *Megasphaera* sp. In addition, butyryl-CoA *L. amylovorus* dehydrogenase – the enzyme involved in the butyryl CoA–acetyl CoA transferase pathway was present in the genome of *Megasphaera* sp. NMBHI-10. In the butyryl CoA–acetyl CoA transferase pathway, H-Ac is metabolized by butyrate-producing bacteria for the synthesis of H-Bu. Therefore, the predominance of *Megasphaera* sp. would explain the increasing levels of H-Bu and decreasing levels of H-Ac observed during microaerobic conditions in this study.

On top of carbohydrate metabolism, the genome of *Megasphera* sp. NMBHI-10 also had several mechanisms for protection against oxidative stress. The presence of enzymes capable of regulating oxidative stress thus allowed the survival of *Megasphaera* sp. under microaerobic conditions. The ability of *Megasphaera* species to survive under oxidative stress and ferment soluble COD into H-Bu correlated well to the negative ORP levels, increasing levels of soluble COD, rising H-Bu levels and falling H-Ac levels observed during microaerobic conditions.

The second predominant group – R1Bac_18 (18%) was closely related to uncultured *Prevotella* sp. (DQ168844). *Prevotella* belongs to phylum *Bacteriodetes* and is a group of asaccharolytic, anaerobic and proteolytic bacteria. Under anaerobic conditions, asaccharolytic bacteria ferment nitrogenous compounds into organic acids and ammonia while proteolytic bacteria degrade nitrogenous compounds into small peptides and amino acids (Takahashi, 2003). During amino acid degradation, *Prevotella* sp. are capable of producing organic acids and ammonia. The base produced contributed to the acid-neutralizing activity and played a role in maintaining pH in the reactor. The predominance of R1Bac_18, which was affiliated to uncultured *Prevotella* sp., would explain the higher levels of NH_3-N observed during microaerobic conditions. The capability of *Prevotella* to neutralize pH and produce organic acids by nitrogenous metabolism would also account for the increasing pH and TVFA levels.

3.2.3. Effect of microaeration on bacteria involved in fermentation of brown water and food waste

An earlier study investigated the microbial community for the anaerobic co-digestion of BW and FW in two-phase CSTR (Lim et al., 2013). As shown in Table 2, there were clear differences in phylogenetic distribution between reactors operated under microaerobic and anaerobic conditions. In comparison to the reactor operated under anaerobic conditions, the bacterial community of the acidogenic reactor operated under microaerobic conditions (in this study) was significantly more diverse. A total of 18 bacterial OTU (classified into 4 phyla) were identified for microaerobic conditions whereas only 7 bacterial OTU (classified into 2 phyla) were identified for anaerobic conditions. In particular, the distribution of bacteria within the phylum *Firmicutes* was higher and that of *Proteobacteria* was lower in the microaerobic reactor. Since *Firmicutes* are an important group of fermentative bacteria that produces extracellular enzymes, the greater distribution of *Firmicutes* in the microaeration reactor led to a higher degree of hydrolysis during microaeration as compared to anaerobic conditions. A comparison of the bacterial communities in this study and that of Lim et al. (2013) showed that the presence of oxygen in the anaerobic digester caused the bacterial community structure to change quite significantly. Microaeration conditions also led to the development of a more diverse bacterial population that will likely enable the acidogenic reactor to metabolize a greater variety of substrates.

Table 2
Comparison of bacterial community in acidogenic reactor operated under microaerobic and anaerobic conditions.

	With microaeration (This study)	No microaeration (Lim et al., 2013)
Bacterial OTU	18	7
Phylum	4	2
Firmicutes	72%	58%
Proteobacteria	2%	42%
Bacteriodetes	24%	0%
Actinobacteria	2%	0%

A study by Tang et al. (2004) investigated the microbial communities of thermophilic digesters treating synthetic FW under anaerobic and microaeration conditions. Similar to this study, the study by Tang et al. (2004) found that microorganisms affiliated with the phylum *Firmicutes* were dominant, independent of the aeration conditions. However, Tang et al. (2004) reported that microaeration did not cause a dramatic shift in the structure of microbial community. On the contrary, this study showed that microaeration has led to obvious effects on the bacterial diversity in the mesophilic digester treating BW and FW. Although *Firmicutes* was predominant in both microaeration conditions (72%) and anaerobic conditions (58%), both the diversity and composition of *Firmicutes* were significantly higher in the microaeration as compared to anaerobic reactor.

4. Conclusion

The bacterial population for microaeration reactor was composed of phyla *Firmicutes*, *Bacteriodetes*, *Proteobacteria* and *Actinobacteria* in proportions of 72%, 24%, 2% and 2% of the bacterial clones, respectively. Many of the bacterial species possessed the ability to consume oxygen and maintain a reducing environment such that fermentation could occur despite higher amounts of oxygen introduced. As compared to anaerobic conditions, microaeration led to a significantly more diverse bacterial community. There was a greater distribution of *Firmicutes*, which enabled the acidogenic reactor to metabolize a greater variety of substrates, giving rise to enhanced COD solubilization and VFA production under microaeration conditions.

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